What is Simulation?

- Simulation is like a pretend game where we mimic how real things work.
- We can do this with pen and paper or using a computer. The idea is to create a fake version of a system and then watch how it behaves to learn about the real system.
- When we study a system over time, we create a simulation model. This model is like a blueprint where we make assumptions about how the system works.
- These assumptions are written down using maths, logic, and symbols to describe the relationships between different parts of the system.
- Once we have our model, we can ask "what if" questions to see how changes might affect the real system. This helps us make decisions, especially when designing new systems or improving existing ones.
- Sometimes, we can solve these models using maths techniques like calculus or probability. This gives us numbers that show how well the system performs.
- But often, real systems are too complicated for this. In those cases, we use computer simulations to imitate how the system behaves over time.
- We collect data from these simulations just like we would if we were observing a real system.
- This data helps us understand how the system works and how different changes might affect its performance.
- By using simulation modelling, we can better understand complex systems and make smarter decisions about them.
- It's like playing with a model of a car before building the real thing, helping us avoid mistakes and improve outcomes.

When Simulation Is the Appropriate Tool

- Simulation has become a widely embraced tool in operations research and systems analysis due to advancements in technology and methodologies.

- Understanding when simulation is the right tool to use is essential for effective problem-solving and decision-making. Here are some circumstances where simulation is appropriate:

1. Studying Complex Systems: Simulation enables the study and experimentation of internal interactions within complex systems or subsystems.
2. Modelling Changes: Informational, organisational, and environmental changes can be simulated to observe their effects on a model's behaviour.
3. System Improvement: Insights gained during the design of a simulation model can suggest improvements for the system under investigation.
4. Variable Analysis: Changing simulation inputs and observing outputs can provide valuable insights into the importance of variables and their interactions.
5. Educational Purposes: Simulation serves as a pedagogical tool to reinforce analytical solution methodologies.
6. Design and Policy Experimentation: Simulation allows experimentation with new designs or policies before implementation, aiding in preparation for potential outcomes.
7. Verification of Analytical Solutions: Simulation can be used to verify analytical solutions obtained through other means.
8. Determining Machine Requirements: Simulating different capabilities for a machine helps in determining its requirements.
9. Training: Simulation models designed for training facilitate learning without the cost and disruption of on-the-job instruction.
10. Visualisation through Animation: Animation of simulated operations allows visualisation of the system, aiding in planning and decision-making.
11. Dealing with Complexity: Modern systems are often too complex to be treated analytically, making simulation necessary for understanding internal interactions.

When Simulation Is Not Appropriate

- While simulation is a powerful tool, there are situations where it may not be the best approach. Here are some guidelines for when simulation might not be appropriate:

1. Common Sense Solutions: Simulation should not be used when the problem can be solved straightforwardly using common sense.
2. Analytical Solutions: If the problem can be solved using analytical methods, simulation may not be necessary.
3. Cost Considerations: If direct experiments are less expensive than simulation and provide adequate results, simulation may not be justified.
4. Costs vs. Savings: If the costs of conducting a simulation exceed potential savings, simulation may not be appropriate.
5. Resource and Time Constraints: Simulation should not be pursued if there are insufficient resources or time available to complete the study.
6. Data Availability: If necessary data for simulation is unavailable, simulation may not be advisable.
7. Verification and Validation Challenges: If there are challenges in verifying and validating the simulation model, simulation may not be suitable.
8. Unrealistic Expectations: If there are unreasonable expectations or an overestimation of the power of simulation, it may not be the right tool.
9. Complex System Behavior: If system behaviour is too complex or cannot be accurately defined, simulation may not be appropriate, especially when dealing with human behaviour.

Advantages and Disadvantages of Simulation

- Simulation is a powerful tool in problem-solving and decision-making, as it closely mimics real-world scenarios, providing insights that are valuable for both existing systems and those in the design stage.
- However, like any method, it has its advantages and disadvantages.

Advantages:

1. Exploration without Disruption: New policies, procedures, or designs can be explored without disrupting ongoing operations of the real system.
2. Risk-Free Testing: New hardware designs or layouts can be tested without committing resources for their acquisition, minimising risk.
3. Feasibility Testing: Hypotheses about certain phenomena can be tested for feasibility, providing valuable insights into system behaviour.
4. Time Manipulation: Time can be compressed or expanded to observe system behaviour at different speeds, facilitating analysis.
5. Variable Interaction Insight: Simulation provides insight into the interaction of variables within the system, aiding in understanding complex relationships.
6. Variable Importance Identification: It helps in understanding the importance of variables to the system's performance, guiding decision-making.
7. Bottleneck Analysis: Bottlenecks can be identified to discover areas where delays occur excessively, facilitating process optimization.
8. Operational Understanding: Simulation helps in understanding how the system operates in reality, rather than how individuals perceive it, leading to more informed decisions.
9. Scenario Analysis: "What if" questions can be answered, particularly useful in designing new systems, allowing exploration of various scenarios.

Disadvantages:

1. Specialised Training: Model building requires specialised training and experience, making it an art that develops over time.
2. Interpretation Challenges: Simulation results can be difficult to interpret due to randomness, making it hard to distinguish between system interrelationships and random fluctuations.
3. Resource Intensive: Simulation modelling and analysis can be time-consuming and expensive, requiring sufficient resources to ensure adequacy.

4. Over-reliance on Simulation: Simulation is sometimes used when an analytical solution is possible or preferable, leading to unnecessary complexity and cost.

- In defence of simulation, advancements in simulation software, output-analysis capabilities, and faster computing have mitigated some of these disadvantages.
- Additionally, closed-form models (a model for which a solution can be obtained using algebraic or calculus techniques, without the need for iterative computation or approximation) often cannot analyse the complexity of real-world systems encountered in practice, making simulation a valuable tool despite its challenges.